BRG-ISEA Lab Activities

Session 1a: Setting Up Linux

- Create a GitHub account and start documenting progress.
    - → Go to https://github.com and sign up for a new account.
    - → Create a new repository (e.g., BRG-27-labs).
    - → Clone the repository to your local machine using Git.
    - → Add a README.md and start documenting each lab you complete.
- Lab: Obtaining Linux on your PC – Install Ubuntu using VirtualBox.
    - → Download Ubuntu ISO from the official website.
    - → Install VirtualBox and create a new virtual machine.
    - → Configure VM settings (e.g., 4GB RAM, 20GB disk).
    - → Mount ISO and install Ubuntu through guided installer.
- Lab: Familiarity with Ubuntu Linux – Basic command line navigation and utilities.
    - → Practice using `pwd`, `ls`, `cd`, `mkdir`, and `touch`.
    - → Understand directory structure (`/etc`, `/var`, `/home`).
    - → Use `man` to explore Linux manual pages.

Session 1b: Exploring Linux

- Lab: Linux Services – Understanding and managing background services.
    - → List services using `systemctl list-units --type=service`.
    - → Start/stop services with `sudo systemctl start|stop [service]`.
    - → Check status of services using `sudo systemctl status`.
- Lab: Linux Permissions – Explore and apply file and directory permissions.
    - → Use `ls -l` to view permissions.
    - → Change permissions using `chmod` (e.g., `chmod 755 file.sh`).
    - → Change ownership using `chown user:group file.txt`.
- Lab: Searching Filesystems – Use commands like `find` and `grep`.
    - → Use `find /path -name 'filename'` to locate files.
    - → Use `grep -r 'search-term' /path/` to search content.

Please remember to capture screenshots or outputs of your lab activities for each session. These are essential for your GitHub documentation and final demonstration.

Session 2a: Total Cost of Ownership

- Lab: Total Cost of Ownership – Apply TCO concepts in practical comparison.
    - → Compare cloud vs on-prem cost (hardware, software, licensing).
    - → Use Excel or Google Sheets to document monthly and yearly costs.
    - → Calculate break-even point, ROI and 3-year projections.
    - → Compare TCO across cloud providers

Session 2b: Cloud Services

- Lab: Cloud Computing – Launch and configure EC2 instance or Azure VM.
    - → Create free tier account on AWS or Azure.
    - → Launch Ubuntu instance (t2.micro or B1s).
    - → Configure SSH keypair, security groups/firewall.
    - → SSH into VM and update OS packages.
- Lab: Bash Scripting – Write simple shell scripts for server automation.
    - → Create a `.sh` file and add `#!/bin/bash` at top.
    - → Use `echo`, `if`, `for`, `while`, `cron` examples.
    - → Run scripts using `bash script.sh` or `chmod +x script.sh`.

Session 3a: DNS & Certificates

- Lab: DNS – Configure and test DNS using BIND or cloud provider.
    - → Register a domain (or use free domain service like Freenom).
    - → Configure A record to point to server public IP.
    - → Use `dig` or `nslookup` to verify DNS propagation.
- Lab: Digital Certificates – Use Let's Encrypt to secure a server.
    - → Install Certbot: `sudo apt install certbot python3-certbot-nginx`.
    - → Run `sudo certbot --nginx` or `--apache`.
    - → Verify certificate using `https://yourdomain` in browser.

Session 3b: Server Automation

- Lab: Scripting Linux Server Functions – Automate server tasks using scripts.
    - → Write script to automate update/backup tasks.
    - → Use cron jobs: `crontab -e` to schedule the script.
    - → Log output to a file using `>> /var/log/task.log`.

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Sessin 4a: Lab Consultation & Additional Server Services

- Participated in Q&A session with teaching staff
  - → List issues you've encountered and bring them up.
  - → Test all configurations before final presentation.
  - → Cloud First and Server Architecture Design – Considerations & Approaches
- Experiment with another server service of interest (in discussion with lecturer)
  Installed, configured and tested an additional service of your choice.
  - → Choose from: MySQL, Jenkins, Samba, FTP, Docker, etc.
  - → Download source for server service to be installed
  - → Install service eg. using apt or curl.
  - → Configure and test the basic functionality.
  - → Documented service configuration and testing steps in GitHub

Session 4b: Lab documentation & Reflection Journal

- Lab activity documentation and peer feedback.
  - → Prepare GitHub README with clear documentation.
  - → Create screenshots of your servers and commands.
  - → Be ready to explain what you did and what you learnt.
  - → Prepared a list of unresolved issues or conceptual doubts
  - → Received final feedback and clarifications on your server deployment
  - → Practised articulating technical concepts as a consultant would
- Reflection on lab consultancy, peer demonstrations and self-assessment
  - → Noted key improvements, obstacles faced, and how they were resolved
  - → Tools you used, Insights gained, lessons learned
  - → Related experience to future IT roles (System Admin, DevOps, etc.)

Video Demonstration Requirements

- Demonstration of all completed labs and your working cloud/Linux server.
- Self-narration explaining what you learned, including scripts, commands, and outputs.
- Webcam view included, with verbal reflection on challenges, insights, and decisions.
- Walkthrough of the additional server service you explored.
- Video link provided must be downloadable

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